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# Trajectory and Power Control for Sustainable UAV-Assisted NOMA-Enabled Backscattering IoT †

**Tianyi Zhang, Mengqin Gu, Deepak Mishra *, Jinhong Yuan and Aruna Seneviratne**

School of Electrical Engineering and Telecommunications, University of New South Wales (UNSW), Sydney, NSW 2052, Australia; tianyi.zhang2@student.unsw.edu.au (T.Z.); m.gu@unsw.edu.au (M.G.); j.yuan@unsw.edu.au (J.Y.); a.seneviratne@unsw.edu.au (A.S.)

* Correspondence: d.mishra@unsw.edu.au

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## Highlights

**What are the main findings?**

- A joint optimization framework for UAV-assisted NOMA-enabled backscatter communication is developed, where closed-form solutions for reflection coefficients and collection locations, KKT-based globally optimal power allocation, and a genetic algorithm-based trajectory design are derived to maximize the sum-throughput rate.
- The proposed joint design achieves an overall performance enhancement of 8.983 dB compared to benchmark schemes that optimize only individual variables, with power allocation identified as the most influential factor contributing 8.295 dB improvement.

**What are the implications of the main findings?**

- The proposed framework demonstrates that integrating UAV mobility with NOMA can effectively overcome the coverage and spectral efficiency limitations of backscatter communication, enabling scalable data collection from large-scale battery-less IoT networks.
- The closed-form and globally optimal solutions offer low computational complexity suitable for practical deployment, providing design guidelines for energy-efficient UAV-assisted green IoT systems.

## Abstract

As mobile networks increasingly support sustainable and green Internet of Things (IoT) applications, energy-efficient solutions that address coverage constraints have become paramount. Although backscatter communication (BackCom) offers a low-power option for IoT devices, particularly battery-less IoT nodes, it can suffer from limited coverage. To overcome this, we exploit aerial platforms (UAVs) integrated with non-orthogonal multiple access (NOMA) to enhance both coverage and spectral efficiency. In this paper, we propose a UAV-supported NOMA-enabled BackCom system to serve massive backscatter node (BN) networks. We aim to maximize system throughput by jointly optimizing the power allocation and reflection coefficients of the BNs, along with the trajectory and data collection locations of the UAV. We derive closed-form solutions for the reflection coefficients and the optimal collection locations of the UAV and achieve global optimality in power allocation by utilizing the Karush–Kuhn–Tucker (KKT) optimality conditions in conjunction with the golden-section search (GSS). In addition, we formulate the UAV trajectory optimization problem as a Traveling Salesman Problem (TSP) and propose an efficient low-complexity genetic algorithm (GA)-based solution. The numerical results demonstrate that the proposed scheme outperforms the benchmark schemes in terms

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of sum-throughput rate and achieves an overall performance enhancement of 8.983 dB, underscoring the potential of our approach for large-scale battery-less IoT deployments.

**Keywords:** backscatter communication; IoT; NOMA; power control; protocol; trajectory design; UAV

## 1. Introduction

Backscatter communication (BackCom) has been recognized as an effective solution for enabling green communication and supporting the Internet of Things (IoT) due to its low power consumption and cost-effective design [1]. In particular, BackCom is well suited for battery-less IoT devices, which can operate solely by harvesting ambient energy. However, BackCom systems are often limited by the unpredictability and availability of ambient RF sources for passive tags, which constrains their data transmission capabilities compared to conventional wireless networks [2]. Despite these limitations, BackCom is regarded as a critical component in future IoT applications, providing low-latency and energy-efficient communication between devices [3]. More broadly, intelligent communication frameworks that leverage sensing-assisted adaptive data transmission have also been explored to reduce spectrum consumption in edge IoT applications [4]. To overcome these challenges, unmanned aerial vehicles (UAVs) can be deployed to maintain a line-of-sight (LoS) connection with ground-based tags, enhancing communication efficiency, coverage, and overall network performance [5]. The aerial flexibility of UAVs allows them to extend the communication range and improve the performance of remote devices, making them an effective medium for augmenting terrestrial wireless networks [6]. UAV-assisted BackCom networks offer mobility, agility, and resilience while being more energy-efficient than traditional methods [7,8]. However, the operation of UAVs is constrained by limited onboard energy, necessitating frequent battery recharges and efficient energy management, which poses a critical challenge in practical deployments. These challenges are exacerbated in multi-UAV coordination scenarios, where conventional time division multiple access (TDMA) protocols may lead to inefficient resource utilization in dynamic environments with rapidly moving UAVs. TDMA also struggles to simultaneously serve a large number of passive backscatter devices, resulting in poor scalability and degraded network performance. In large-scale IoT deployments with many battery-less tags, the shortage of orthogonal resources in TDMA-based schemes further limits achievable spectral efficiency [9]. To address these issues, integrating non-orthogonal multiple access (NOMA) into UAV-assisted BackCom networks has been proposed as an effective strategy to enhance spectral efficiency and connectivity [10]. NOMA, implemented through power-domain or code-domain schemes, allocates transmission power among users and employs successive interference cancellation (SIC) at receivers to enable simultaneous access for multiple devices. This not only improves spectrum utilization but also reduces access latency, rendering it particularly applicable to densely deployed, large-scale IoT environments. Motivated by these benefits, this paper investigates UAV-assisted NOMA backscattering to achieve energy-efficient, reliable coverage, thereby enhancing the spectral efficiency of UAV-assisted NOMA systems.

### 1.1. State of the Art

Recently, several studies have explored UAV backscattering. The trade-off between outage probability and UAV data collection location is examined in [11], where the optimal location is determined to achieve maximum energy efficiency while minimizing the UAV transmit power. Yang et al. [12] propose an iterative method to simultaneously optimize

drone trajectory, tag scheduling, and transmission power, aiming to maximize energy efficiency (EE) under energy throughput and harvested energy constraints. Height-related tasks and sensor activation are addressed in [13], where the authors minimize errors and reconstruct both the elevation angle and trajectory. Moreover, Hua et al. [14] focus on optimizing the reflection coefficient and time allocation under fixed ambient constraints to maximize ergodic capacity. In addition, the UAV trajectory planning problem for age-of-information (AoI)-minimal data collection in IoT networks has been formulated as a generalized TSP and solved using learning-based approaches such as the transformer-weighted-A algorithm [15], highlighting the growing interest in AoI-aware intelligent trajectory design for UAV-assisted IoT systems. Furthermore, the NOMA technique significantly contributes to improving spectral efficiency. The interrelationship between IoT and cellular networks, with and without NOMA, is demonstrated through numerical results in [16]. Bit error rate and reflection coefficient evaluations are conducted in [17]. Khan et al. [18] show that jointly designing the backscatter reflection coefficient and transmit power of IoT users achieves superior performance compared to both pure NOMA and conventional OMA approaches. Resource allocation is explored in NOMA-enhanced BackCom systems [19]. UAV also offers substantial support for NOMA, as evidenced in [20], where the optimization of NOMA base station precoding vectors is employed to achieve sum rate maximization in UAV networks. The UAV-assisted backscatter NOMA scheme further enhances efficiency and reliability, as demonstrated in [21,22]. The optimal reflection coefficient and UAV location at a fixed altitude are determined using a block-coordinate descent method with quadratic transform techniques in [21]; however, ref. [21] does not consider trajectory optimization, power allocation, or UAV energy consumption, and is limited to a small-scale deployment. Meanwhile, the UAV altitude is optimized to maximize the throughput-to-flight-time ratio in [22], where the reflection coefficients are pre-assigned rather than optimized, and neither power allocation nor collection location design is addressed. In contrast to [21,22], our approach jointly optimizes the reflection coefficients, collection locations, transmit power allocation, and UAV trajectory under a practical energy consumption constraint and is validated for large-scale massive BN deployments. To achieve this, we decompose the complex joint optimization problem into convex subproblems and apply an iterative optimization algorithm that ensures convergence to a stationary solution.

### *1.2. Motivation and Contribution*

Based on the aforementioned discussion, BackCom can cover wide communication ranges based on tag distribution, while UAVs represent the best utilization of tag-excitation, and NOMA is key to pairing. Therefore, the combination of UAV-assisted NOMA BackCom creates a sustainable, green communication network with broad applications. Although there have been a few studies on UAV-assisted NOMA backscatter networks, this area still presents a research gap. Motivated by the practicality and applicability of the technology, we propose a comprehensive scheme in this paper to improve the sum rate by scheduling UAVs and optimizing data transfer. It is worth noting that this paper is a substantial extension of our preliminary conference work [23], which focused on pair-wise hovering location selection and power control based on a randomly assigned UAV trajectory. In contrast, this journal version introduces several major extensions, including the joint optimization of the UAV trajectory and backscattering reflection coefficients, leading to new non-convex subproblems with derived optimal solutions; a refined joint design strategy with a new iterative algorithm and a detailed computational complexity analysis; significantly expanded literature review, numerical results, and detailed mathematical derivations and proofs; additional analytical insights on energy harvesting and energy-

awareness; as well as new and updated figures offering deeper insights into the proposed framework. The five-fold contribution of this paper is outlined as follows:

- We propose a novel NOMA-enabled UAV-aided monostatic BackCom system framework to efficiently serve large-scale backscatter node (BN) networks. Building upon this model, we develop a NOMA-enabled backscattering protocol for the two-user NOMA scheme that dynamically adjusts the reflection coefficients of the backscatter nodes, effectively orchestrating user pairing and power control in each time slot to achieve robust data collection and high spectral efficiency.
- We formulate a joint optimization problem for the adopted NOMA BackCom model to maximize the sum-throughput by jointly optimizing the BN reflection coefficients, UAV trajectory, transmit power allocation, and data collection locations, under a practical UAV energy-consumption constraint.
- Since the formulated joint optimization problem is inherently non-convex, we decompose it into four tractable convex subproblems. Closed-form solutions are derived for the reflection coefficients and the UAV's optimal collection locations, and global optimality in power allocation is guaranteed by applying the Karush–Kuhn–Tucker (KKT) optimality conditions along with the golden-section search (GSS) method.
- The UAV trajectory optimization problem is modeled as a Traveling Salesman Problem (TSP), and a low-complexity genetic algorithm (GA)-based method is developed to yield near-optimal trajectories at a substantially lower computational cost.
- Numerical analyses are conducted to validate key analytical claims, provide insight into the optimal joint design, and demonstrate the effectiveness of our optimization approach in enhancing system throughput, leading to significant performance gains over benchmark schemes.

### *1.3. Organization and Notation*

The remaining part of the paper is structured as follows. The NOMA-enabled UAV-assisted BackCom system model is proposed in Section 2, and the problem formulation is detailed in Section 3. The proposed solution methodology is discussed in Section 4, while the numerical simulation results are provided in Section 5. Finally, the concluding remarks and future directions are outlined in Section 6.

Next, we summarize the notations used in this paper. Vectors and matrices are denoted by boldface lowercase and boldface uppercase letters, respectively. The $\mathbb{E}\{\cdot\}$ denotes the expectation operator. $0_{N\times 1}$ and $\mathbf{I}_N$ denote the $N \times 1$ zero vector and the $N \times N$ identity matrix, respectively. $\mathbf{v}_{\max}\{\mathbf{G}\}$ denotes the eigenvector associated with the largest eigenvalue of $\mathbf{G}$. $\mathbb{CN}(\mu, \sigma)$ represents the circularly symmetric complex Gaussian (CSCG) distribution with mean $\mu$ and variance $\sigma$. $\mathrm{Tr}(\mathbf{A})$ represents the trace of a matrix. Finally, in this paper, the used variables are summarized in Table 1.

**Table 1.** Used variables with their descriptions.

| Variables | Descriptions |
|---|---|
| $N$ | Number of Slots |
| $K$ | Number of BNs |
| $n$ | Slot Sequence Number |
| $(x_i[n], y_i[n])$ | Location of the $i$th BN |
| $(x_u[n], y_u[n])$ | Data Collection Location of the UAV |
| $z$ | Altitude of the UAV |
| $\phi_i[n]$ | Elevation Angle at the $i$th BN |

**Table 1.** *Cont.*

| Variables | Descriptions |
|---|---|
| $\beta_0$ | Channel Power Gain |
| $\kappa$ | Attenuation Factor |
| $\Gamma_i[n]$ | Reflection Coefficient for the $i$-th BN |
| $L_i[n]$ | Average Path Loss at the $i$-th BN |
| $P_t[n]$ | Transmitted Power in the $n$-th slot |
| $P_{tot}$ | Total Transmitted Power |
| $\vartheta_u$ | Speed of UAV |
| $\mathcal{T}_P$ | Thrust Power of UAV |
| $\mathcal{T}$ | Potential Routes of UAV |
| $P_f$ | Propulsion Power of UAV |
| $t_{hov}$ | Hovering Time for Data Collection |
| $E_f$ | Flying Energy Consumption |
| $E_t$ | Total Transmit Energy |
| $D_f$ | Total Flying Distance |
| $E_{max}$ | Total Energy Consumption Budget |
| $P_b$ | Total Transmitted Power Budget |

## 2. System Description

We propose a NOMA-enabled UAV-assisted BackCom network designed for massive data collection from $K$ BNs, which are randomly and independently distributed within a designated area, as illustrated in Figure 1a. Utilizing a two-user NOMA scheme, the network organizes the $K$ BNs into $K/2$ pairs to form a sensing network. A quad-rotor UAV, equipped with a single antenna, collects data from each pair of BNs during a collection time $t_{hov}$. Operating at a fixed altitude, the UAV follows a predetermined trajectory that begins and ends at specific points, which may be identical or distinct. Along this trajectory, the UAV hovers at $N$ discrete points, collecting data from each pair of BNs during its hover period. Between hovering points, the UAV travels along a straight-line path with constant velocity, ensuring efficient transitions and optimized data transmission.

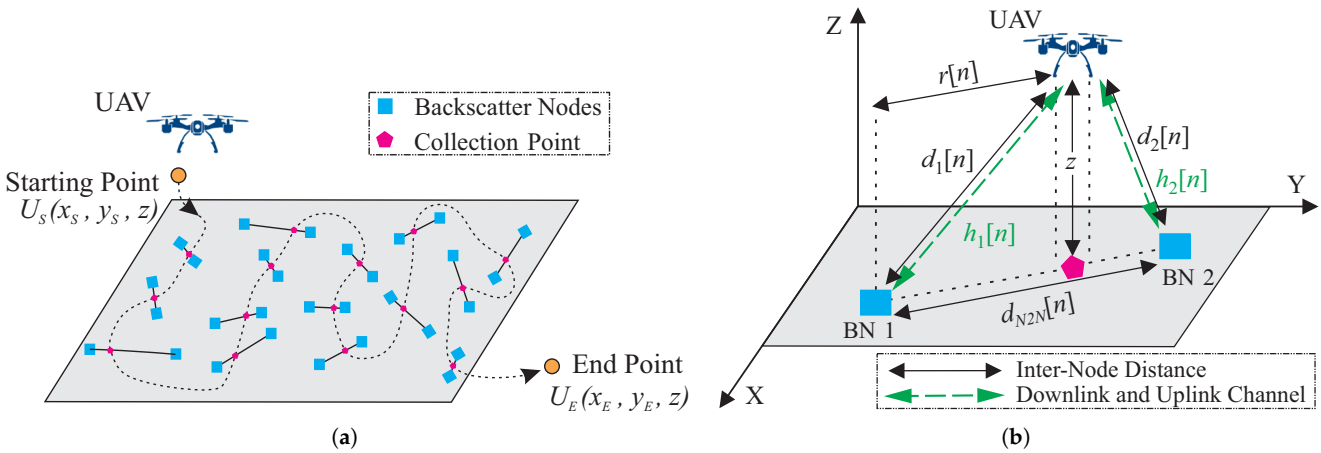



**Figure 1.** The proposed NOMA-enabled UAV-assisted BackCom system networks: (**a**) The UAV flight trajectory overview, and (**b**) Data collection from a pair of BNs during a single time slot.

### 2.1. Network Topology and Wireless Channel Model

The UAV follows a two-dimensional (2D) flight trajectory at a fixed altitude $z$ with a uniform speed $\vartheta_u$. In each time slot, a pair of BNs is served simultaneously, which necessitates a total of $N$ time slots to accommodate all pairs, with $N = \frac{K}{2} \in \mathcal{N} \triangleq \{1, 2, \ldots, n, \ldots, N\}$. The UAV's hovering position projected onto the ground in each slot is represented by the coordinates $\mathbf{u}[n] = (x_u[n], y_u[n])$, while the BN positions in a single time slot are denoted by $(x_i[n], y_i[n])$, where $i \in \{1, 2\}$ refers to the specific BN. The node-to-node (N2N) distance between two BNs in a pair during each time slot can be expressed as $d_{N2N}[n] = \sqrt{(x_2[n] - x_1[n])^2 + (y_2[n] - y_1[n])^2}$. The two-user NOMA BackCom system model is illustrated in Figure 1b. Since this scenario emulates a practical deployment, the air-to-ground (A2G) channel measurements are affected by both path loss and shadowing effects. To capture these propagation characteristics, both LoS and non-line-of-sight (NLoS) components are incorporated due to the potential presence of obstacles. Let $r[n]$ denote the lateral distance between the UAV and the first BN in the $n$th slot, such that the distance between the first BN and the UAV is given by $d_1[n] = \sqrt{r[n]^2 + z^2}$. Accordingly, the distance between the second BN and the UAV can be expressed as $d_2[n] = \sqrt{(d_{N2N}[n] - r[n])^2 + z^2}$. Consequently, based on [24], by adopting the free-space path loss model, the path loss between the UAV and BNs under both LoS and NLoS conditions is modeled as follows:

$$L_i^{LoS}[n] = \beta_0 \, d_i[n]^{-2}, \quad \forall i \in \{1, 2\}, \tag{1}$$

$$L_i^{NLoS}[n] = \kappa \beta_0 \, d_i[n]^{-2}, \quad \forall i \in \{1, 2\}, \tag{2}$$

where $\beta_0$ denotes the channel power gain at a reference distance of 1 m and $\kappa$ represents the attenuation factor. Accordingly, the probability of LoS path loss is given by [25] as

$$\mathcal{P}_i^{LoS}[n] = \frac{1}{1 + a \exp[-b(\phi_i[n] - a)]}, \qquad \text{and} \tag{3}$$

$$\phi_i[n] = \frac{180}{\pi} \sin^{-1}\left(\frac{z}{d_i[n]}\right), \qquad \forall i \in \{1, 2\} \tag{4}$$

where $a$ and $b$ are environment-dependent constants [26,27] and $\phi_i[n]$ represents the elevation angle with respect to the $i$th BN.

Hence, the average path loss can be expressed as follows:

$$L_i[n] = \mathcal{P}_i^{LoS}[n] L_i^{LoS}[n] + (1 - \mathcal{P}_i^{LoS}[n]) L_i^{NLoS}[n], \tag{5}$$

where the relationship between the LoS and the NLoS probabilities can be written as $\mathcal{P}^{NLoS} = 1 - \mathcal{P}^{LoS}$.

### 2.2. Uplink NOMA-Enabled Backscattering Protocol

The UAV downlinks unmodulated RF signals to the terrestrial BN and retrieves the data backscattered from it. Assuming that the transmitted power $P_t[n]$ remains identical for each time slot and each BN, the received signal $\mathrm{y}_i[n]$ at the $i$th BN during the $n$th slot can be expressed as

$$\mathrm{y}_i[n] = \sqrt{P_t[n]} \, \mathrm{h}_i[n] \, c[n] + n_i, \quad \forall i \in \{1, 2\}, \tag{6}$$

where $\mathrm{h}_i[n] \sim \mathbb{CN}(0, 1)$ represents the small-scale Rayleigh fading coefficient from the UAV to the $i$th BN and $c[n]$ is the carrier signal with unit power, i.e., $\mathbb{E}\{|c[n]|^2\} = 1$. Here, $n_i \sim \mathbb{CN}(0, \sigma_{n_i}^2)$ represents the complex zero-mean additive white Gaussian noise (AWGN) at the $i$th BN. Under an uplink NOMA scheme, two BNs simultaneously backscatter their

modulated signals to the UAV. Accordingly, the retrieved signal $y_u[n]$ at the UAV during the $n$th slot is given by

$$\begin{aligned} y_u[n] &= \sum_{i=1}^{2} \sqrt{\Gamma_i[n]}\, h_i[n]\, y_i[n] + n_u, \\ &\approx \sum_{i=1}^{2} \sqrt{\Gamma_i[n]}\, h_i[n] \sqrt{P_t[n]}\, h_i[n]\, c[n] + n_u, \end{aligned} \tag{7}$$

where $\Gamma_i[n]$ denotes the reflection coefficient for the $i$th BN in the $n$th slot, representing the ratio of the backscattered signal power to the transmitted signal power. Here, the backscattered noise term associated with $n_i$ is practically negligible relative to the excitation power, since it is substantially attenuated during the backscattering process.

In the uplink NOMA scheme, the SIC technique significantly improves decoding efficiency by allowing the receiver to first recover the strongest signal while treating weaker signals as interference. Accordingly, the UAV employs SIC to separate backscattered signals from BN pairs. The optimal SIC decoding order follows a descending order according to the users' channel gain [28], which is characterized by $|h_i[n]|^2$. Based on the measured channel gains, we designate the node with the strongest channel as the second BN under the condition $|h_2[n]|^2 > |h_1[n]|^2$. Consequently, the first BN can be treated as interference. Thus, the UAV initially decodes the signal of the second BN by regarding the first BN's signal as noise. Afterward, the UAV subtracts the decoded signal of the second BN from the received signal to recover the first BN's signal. The achievable throughput of each BN is expressed as

$$R_1[n] = \log_2\left(1 + \frac{\Gamma_1[n] P_t[n] |h_1[n]|^4 |L_1[n]|^2}{\sigma_{n_u}^2}\right), \tag{8}$$

$$R_2[n] = \log_2\left(1 + \frac{\Gamma_2[n] P_t[n] |h_2[n]|^4 |L_2[n]|^2}{\Gamma_1[n] P_t[n] |h_1[n]|^4 |L_1[n]|^2 + \sigma_{n_u}^2}\right). \tag{9}$$

### *2.3. Energy Consumption Model of Quad-Rotor UAV*

Since green communication is targeted, reducing the energy consumption of the UAV is of critical importance. In our scenario, a quad-rotor UAV model is adopted, and its energy consumption is analyzed from two aspects. First, the UAV requires propulsion power to generate thrust, counteracting both gravitational and drag forces to enable mobility. Second, energy is expended for data transmission during the collection process. In addition, a relatively small amount of energy is consumed by on-board circuitry, such as signal processing, remote control communication, and other electronic components. However, in practice, this circuitry-related energy is generally two orders of magnitude lower than the UAV's propulsion energy.

Therefore, in this particular scenario, we focus solely on the propulsion and transmission energy of the quad-rotor UAV. For clarity, we assume that the UAV flies in a straight line between hovering points in each time slot at a constant speed (the constant speed assumption during inter-waypoint flight segments is widely adopted in the UAV-assisted communications literature [11,13,14,20,29], where the primary focus is on communication system optimization rather than detailed flight dynamics modeling) and altitude. Since the UAV cruises at a constant speed in the horizontal plane, the forward acceleration along the flight direction is zero. Similarly, since the UAV operates at a fixed altitude, the vertical acceleration and climb angle are both zero. Direction changes between consecutive waypoints are modeled through rotor control adjustments, and the associated transient maneuvering cost is neglected for tractability, consistent with the standard constant-speed

rotary-wing communication model adopted in this work [11,13,14,20,29]. Under normal operating conditions, the thrust generated by each rotor can be expressed as a function similar in [30], given by

$$\mathcal{T}_P[\vartheta_u] = \frac{1}{n_r}\left(\frac{1}{2}\rho\,\vartheta_u^2\,S_{FP}\,\vartheta_d\;-\;m\mathrm{g}\right), \tag{10}$$

where $n_r$ denotes the number of rotors; $\rho$ and $S_{FP}$ represent the air density and the fuselage equivalent flat plate area, respectively; $m$ is the mass of the UAV, and g is the gravitational acceleration vector; $\vartheta_u$ and $\vartheta_d$ denote the forward velocity of the quad-rotor UAV and the velocity direction vector, respectively. Accordingly, the required propulsion power $P_f$ for a quad-rotor UAV operating at constant speed can be formulated as (11), shown at the bottom of the page, where $\delta$ is the local blade section drag coefficient; $c_T$ denotes the thrust coefficient based on the disc area; $c_f$ represents the incremental correction factor of the induced power; $d_0$ is the fuselage drag ratio per rotor; finally, $A$ and $c_s$ denote the disc area per rotor and the rotor solidity, respectively. The corresponding parameters are summarized in Table 2. (The propulsion and aerodynamic parameters in this table are adopted from the generalized rotary-wing UAV model in [30], based on theoretical quad-rotor dynamics modeling. Since commercial UAV datasheets typically do not disclose aerodynamic coefficients, a strict parameter-level mapping to a specific commercial UAV model is not feasible. Nonetheless, the key kinematic parameters, such as the takeoff mass and cruising speed, are broadly consistent with lightweight commercial quad-rotor platforms such as the DJI Phantom 4 and DJI Matrice 100, confirming the practical relevance of the adopted parameter set.).

$$P_f[\vartheta_u] = n_r\left[\frac{\delta}{8}\left(\frac{\mathcal{T}_P[\vartheta_u]}{c_T\rho A}+3\vartheta_u^2\right)\sqrt{\frac{\mathcal{T}_P[\vartheta_u]\,\rho\,c_s^2\,A}{c_T}}\;+\;\left(1+c_f\right)\mathcal{T}_P[\vartheta_u]\left(\sqrt{\frac{\mathcal{T}_P^2[\vartheta_u]}{4\rho^2A^2}+\frac{\vartheta_u^4}{4}}-\frac{\vartheta_u^2}{2}\right)^{\frac{1}{2}}+\;\frac{1}{2}\,d_0\,\vartheta_u^3\,\rho\,c_s\,A\right] \tag{11}$$

**Table 2.** Parameters utilized for the quad-rotor UAV [30].

| Parameter | Value | Parameter | Value |
|---|---|---|---|
| $n_r$ | 4 | $\rho$ | 1.293 kg/m$^3$ |
| $S_{FP}$ | 0.01 m$^2$ | $m$ | 2 kg |
| g | 9.8 m/s$^2$ | $\vartheta_u$ | 15.521 m/s |
| $c_s$ | 0.0955 | $\delta$ | 0.012 |
| $c_T$ | 0.302 | $c_f$ | 0.131 |
| $d_0$ | 0.8333 | $A$ | 0.0314 m$^2$ |

Subsequently, the total energy consumption of the quad-rotor UAV consists of two components, which are the flying propulsion energy $E_f$ and the data transmission energy $E_t$. Their sum must not exceed the total energy consumption budget $E_{max}$, giving rise to the following constraint:

$$E_f\;+\;E_t\;\leq\;E_{max}. \tag{12}$$

Here, we can further define the total transmit energy as $E_t \triangleq \left(\sum_{n=1}^{N} P_t[n]\,t_{hov}\right)$ and the flying energy consumption as $E_f \triangleq \frac{D_f\,P_f}{\vartheta_u}$, where $D_f$ represents the total flying distance. Specifically, the UAV starts at $\mathbf{u}[0] = \mathbf{U_S}(x_S, y_S, z)$ and ends at $\mathbf{u}[N+1] = \mathbf{U_E}(x_E, y_E, z)$.

Therefore, $D_f$ is obtained by accumulating the Euclidean distances between consecutive waypoints, i.e., $D_f \triangleq \sum_{n=0}^{N} ||\mathbf{u}[n+1] - \mathbf{u}[n]||$.

## 3. Problem Formulation

Building upon the analysis in the preceding section, the path loss is defined in Section 2.1 and the received power in Section 2.2. Referring to (8) and (9), the total throughput in the $n$th slot is given by $R[n] = R_1[n] + R_2[n]$. Hence, the sum-throughput $R_{\text{Sum}}$ throughout the flight period can be written as

$$R_{\text{Sum}} = \sum_{n=1}^{N} R[n]. \tag{13}$$

Based on the above, the optimization problem for maximizing the sum-throughput rate is formulated as follows:

$$\begin{aligned}
\mathcal{O}_{\text{S}}: \quad & \max_{\{\mathbf{u}[n], \Gamma_1[n], \Gamma_2[n], r[n], P_t[n],\, \forall n \in \mathcal{N}\}} \quad R_{\text{Sum}}, \\
\text{subject to:} \quad & (\text{C1}): \; 0 \le \Gamma_1[n] \le 1, \; \forall n, \\
& (\text{C2}): \; 0 \le \Gamma_2[n] \le 1, \; \forall n, \\
& (\text{C3}): \; 0 \le r[n] \le d_{N2N}[n], \; \forall n, \\
& (\text{C4}): \; E_f + \left( \sum_{n=1}^{N} P_t[n]\, t_{hov} \right) \le E_{max}, \\
& (\text{C5}): \; \mathbf{u}[0] = \mathbf{U_S}, \\
& (\text{C6}): \; \mathbf{u}[N+1] = \mathbf{U_E}, \\
& (\text{C7}): \; \sum_{n=0}^{N} ||\mathbf{u}[n+1] - \mathbf{u}[n]|| \le D_{max}.
\end{aligned}$$

Here, constraints (C1)–(C2) ensures that the reflection coefficients lie within the valid range. Constraint (C3) ensures the collection point is within the lateral distance parallel to the line segment connecting the two BNs, while (C4) limits the UAV's total energy consumption, including flight and data transmission, to a fixed energy budget. Constraints (C5)–(C6) specify that the UAV's trajectory begins at a fixed starting point and ends at a fixed ending point. Finally, constraint (C7) restricts the total flying distance of the UAV to be no greater than a specified maximum length.

Due to the coupling of multiple variables, the problem $\mathcal{O}_{\text{S}}$ is inherently non-convex, posing a significant optimization challenge. Subsequently, it may be noted that $\mathcal{O}_S$ is NP-hard since the trajectory design subproblem, obtained by fixing all other optimization variables, reduces to a TSP, which is a well-known NP-complete problem that cannot be solved in polynomial time [31]. This motivates our use of a heuristic GA-based solution for the trajectory design. In the following sections, we present the proposed methods to tackle this complex optimization problem.

## 4. Proposed Solution Methodology

This section details the proposed solution methodology, where the original problem is decomposed into individual subproblems. The optimal UAV trajectory is first obtained. Subsequently, the reflection coefficients and power allocation scheme are optimized under the assumption that all relevant parameters are available. Finally, we summarize the implementation details and briefly discuss the complexity analysis.

### *4.1. Optimal UAV Trajectory Design*

This section aims to minimize the overall trajectory distance of the UAV while ensuring that each pair of BNs is served exactly once, thereby preventing any repetition in servicing. The complete trajectory starts from the fixed starting point $\mathbf{U_S}$ and stops at the fixed endpoint $\mathbf{U_E}$, as defined by constraints (C5)–(C6). Assuming that the collection locations are known, an efficient route can be determined to visit all collection points. We denote a binary indicator $\Psi_{jk} = 1$ if the UAV flies a route between two collection points $\mathbf{U}_j(\mathbf{u}[j], z)$ and $\mathbf{U}_k(\mathbf{u}[k], z)$, while $\Psi_{jk} = 0$ otherwise. In addition, the segment length between two collection points is defined as $D_{jk}$, i.e., $D_{jk} = \|\mathbf{u}[k] - \mathbf{u}[j]\|$. Constraint (C7) imposes an upper bound on the total travel distance of the UAV, i.e., the sum of the selected segment lengths $\sum_{j,k} D_{jk}\Psi_{jk}$, which effectively restricts the feasible set of admissible inter-point distances. With these definitions, the trajectory design can be cast into a TSP problem, where the collection locations are treated as cities and the UAV is the Traveling Salesman [29], formulated as a subproblem $\mathcal{P}_{\mathrm{A}}$, given by

$$\mathcal{P}_{\mathrm{A}}: \min_{\{\Psi_{jk}\in\{0,1\},\forall j,k\}} \quad \sum_{j=0}^{N}\sum_{k=1}^{N+1} D_{jk}\Psi_{jk}, \qquad \text{subject to:}$$

$$(\mathrm{C8}): \sum_{j=0, j\neq k}^{N} \Psi_{jk} = 1, \quad \forall k \in \{1, 2, \ldots, N+1\},$$

$$(\mathrm{C9}): \sum_{k=1, k\neq j}^{N+1} \Psi_{jk} = 1, \quad \forall j \in \{0, 1, \ldots, N\},$$

$$(\mathrm{C10}): \sum_{j\in\mathcal{S}}\sum_{k\in\mathcal{S}} \Psi_{jk} \leq |\mathcal{S}| - 1, \ \text{for} \begin{cases} S \subset \mathcal{V}, \\ 2 \leq |\mathcal{S}| \leq N-1, \end{cases}$$

Here, $\mathcal{V}$ describes the set of BNs' locations, and $\mathcal{S}$ is a subset such that $\mathcal{S} \subset \mathcal{V}$. The TSP has been widely studied, and several commercial tools are available to solve it. We specifically use the GA algorithm with reasonable complexity because it provides a good trade-off between computational complexity and speed. The GA algorithm is a heuristic optimization method inspired by natural evolutionary processes that is well suited for solving the TSP [32]. A set of potential routes $\mathcal{T}$ is generated, which converges into an efficient route. The GA-based TSP solver is configured with a population size of 1000 and runs for 5000 iterations. A tournament selection scheme with a tournament size of 5 is adopted, where the best individual from each group is preserved and mutated using three operators: inversion, swap, and slide mutations, while one offspring per group is randomly generated to maintain diversity. These parameters are empirically chosen to achieve a favorable trade-off between solution quality and computational cost.

### *4.2. Optimal Backscattering Reflection Coefficient Design*

This section proposes optimal closed-form solutions for the backscattering reflection coefficients $\Gamma_i[n]$, $\forall i \in \{1, 2\}$. By keeping all other optimization variables fixed, the optimal $\Gamma_1[n]$ and $\Gamma_2[n]$ can be obtained slot by slot. We discuss how to obtain the optimal $\Gamma_1[n]$ and $\Gamma_2[n]$ separately below.

4.2.1. Backscattering Reflection Coefficient for the Second BN

From (8) and (9), the objective function in (13) is first rewritten with respect to $\Gamma_2[n]$. Consequently, the original optimization problem $\mathcal{O}_S$ is decoupled into the following subproblem $\mathcal{P}_{B1}$ with respect to $\Gamma_2[n]$, shown as

$$\mathcal{P}_{B1}: \max_{\{\Gamma_2[n], \forall n \in \mathcal{N}\}} \sum_{n=1}^{N} R_1[n] + \log_2(1 + \Gamma_2[n]\mathrm{b}_1[n]) \quad \text{subject to:} \quad (C2).$$

where we denote $\mathrm{b}_1[n] \triangleq \frac{P_t[n]|h_2[n]|^4|L_2[n]|^2}{\Gamma_1[n]P_t[n]|h_1[n]|^4|L_1[n]|^2 + \sigma_{n_u}^2}$ and $R_1[n]$ as constants. It can be readily observed that the optimization objective function is monotonically increasing with respect to $\Gamma_2[n]$, which implies that the maximum sum-throughput is achieved at $\Gamma_2^*[n] = 1$.

4.2.2. Backscattering Reflection Coefficient for the First BN

Similarly, we can re-write the total throughput $\overline{R}[n]$ in (13) at the $n$th slot with respect to $\Gamma_1[n]$, as shown below:

$$\overline{R}[n] \triangleq \log_2(1 + \Gamma_1[n]\mathrm{b}_2[n]) + \log_2\left(1 + \frac{1}{\Gamma_1[n]\mathrm{b}_3[n] + \overline{\sigma}_{n_u}^2}\right), \tag{14}$$

where we can denote the $\mathrm{b}_2[n] \triangleq \frac{P_t[n]|h_1[n]|^4|L_1[n]|^2}{\sigma_{n_u}^2}$, $\mathrm{b}_3[n] \triangleq \frac{P_t[n]|h_1[n]|^4|L_1[n]|^2}{\Gamma_2[n]P_t[n]|h_2[n]|^4|L_2[n]|^2}$, and $\overline{\sigma}_{n_u}^2 \triangleq \frac{\sigma_{n_u}^2}{\Gamma_2[n]P_t[n]|h_2[n]|^4|L_2[n]|^2}$. Subsequently, $\mathcal{O}_S$ is decoupled into the below subproblem $\mathcal{P}_{B2}$ with respect to $\Gamma_1[n]$, shown as:

$$\mathcal{P}_{B2}: \max_{\{\Gamma_1[n], \forall n \in \mathcal{N}\}} \overline{R}_{\mathrm{Sum}} = \sum_{n=1}^{N} \overline{R}[n], \quad \text{subject to:} \quad (C1).$$

Here, it is straightforward to verify that $\overline{R}[n]$ is concave, as its second-order derivative is strictly negative. As a result, the globally optimal solution for $\mathcal{P}_{B2}$ with respect to $\Gamma_1[n]$ that maximizes the sum-throughput rate can be obtained by optimizing on a slot-by-slot basis. Specifically, by taking the derivative of $\overline{R}[n]$ in the $n$th slot with respect to $\Gamma_1[n]$, the resultant backscattering reflection coefficient $\overline{\Gamma}_1[n]$ can be obtained by solving $\frac{\partial \overline{R}[n]}{\partial \Gamma_1[n]} = 0$, yielding:

$$\overline{\Gamma}_1[n] \triangleq \frac{-\mathrm{b}_2[n]\mathrm{b}_3[n]\overline{\sigma}_{n_u}^2 \pm \sqrt{-\mathrm{b}_2[n]\mathrm{b}_3{}^2[n]\left(-\mathrm{b}_3[n] + \mathrm{b}_2[n]\overline{\sigma}_{n_u}^2\right)}}{\mathrm{b}_2[n]\mathrm{b}_3{}^2[n]}. \tag{15}$$

Here, the detailed derivation is provided in Appendix A. Eventually, the globally optimized $\Gamma_1^*[n]$ for $\mathcal{P}_{B2}$ is determined by selecting the candidate from $\{0, 1, \overline{\Gamma}_1[n]\}$ that yields the highest sum-throughput rate while satisfying constraint (C1).

*4.3. Optimal Power Allocation Design*

In this section, a power allocation scheme is developed to obtain the optimal $P_t[n]$ for maximizing the sum-throughput rate. With all other optimization variables held fixed, the total throughput $\widetilde{R}[n]$ in the $n$th slot with respect to $P_t[n]$ is reformulated as:

$$\widetilde{R}[n] \triangleq \log_2(1 + P_t[n]\mathrm{c}_1[n]) + \log_2\left(1 + \frac{P_t[n]}{P_t[n]\mathrm{c}_2[n] + \widetilde{\sigma}_{n_u}^2}\right), \tag{16}$$

where $\mathrm{c}_1[n] \triangleq \frac{\Gamma_1[n]|h_1[n]|^4|L_1[n]|^2}{\sigma_{n_u}^2}$, $\mathrm{c}_2[n] \triangleq \frac{\Gamma_1[n]|h_1[n]|^4|L_1[n]|^2}{\Gamma_2[n]|h_2[n]|^4|L_2[n]|^2}$, and $\widetilde{\sigma}_{n_u}^2 \triangleq \frac{\sigma_{n_u}^2}{\Gamma_2[n]|h_2[n]|^4|L_2[n]|^2}$. Hence, the sum-throughput rate optimization problem is rewritten as $\mathcal{P}_{\mathrm{C1}}$, given by

$$\mathcal{P}_{\mathrm{C1}}: \max_{\{P_t[n], \forall n \in \mathcal{N}\}} \widetilde{R}_{\mathrm{Sum}} = \sum_{n=1}^{N} \widetilde{R}[n], \quad \text{subject to:} \quad (\mathrm{C4}).$$

**Lemma 1.** *Optimization problem $\mathcal{P}_{\mathrm{C1}}$ is a convex problem.*

**Proof.** Please refer to Appendix B for the proof. □

Noting $\mathcal{P}_{\mathrm{C1}}$ is a convex problem proved by Lemma 1, we can claim that the KKT point is the global optimal solution. Here, the Lagrange multiplier method can be adopted to solve optimization problems like $\mathcal{P}_{\mathrm{C1}}$ subject to constraints [33]. Keeping constraints (C1)–(C3) and (C5)–(C7) implicit, the Lagrangian expression of $\mathcal{P}_{\mathrm{C1}}$ with regard to the constraints (C4) can be expressed as

$$\mathcal{L} \triangleq \sum_{n=1}^{N} \widetilde{R}[n] + \lambda_T \left( E_{max} - E_f - \sum_{n=1}^{N} P_t[n] t_{hov} \right), \tag{17}$$

where $\lambda_T$ represents the Lagrange multiplier associated with (C4). Thus, the optimal power allocation is achieved by solving the respective subgradient KKT conditions (Chapter 5.5.3, [34]).

**Lemma 2.** *The Lagrangian function $\mathcal{L}(P_t[n], \lambda_T)$ is concave with respect to $P_t[n]$ for any fixed Lagrange multiplier $\lambda_T$.*

**Proof.** According to (Equation (3.10), [34]), for a composition function $f = h \circ g$, it is asserted that if $h$ is concave and nondecreasing, while $g$ is concave, then $f$ is concave. Recalling from (16) and (17), the logarithm function can be regarded as $h$, and $g$ as the function of $P_t[n]$ inside the logarithm function. Hence, the Lagrangian function expressed in (17) is concave since both of the logarithm terms are concave with respect to $P_t[n]$. □

$$\frac{\partial \mathcal{L}}{\partial P_t[n]} = -t_{hov}\lambda_T + \frac{\mathrm{c}_1[n]\mathrm{c}_2{}^2[n]P_t^2[n] + \mathrm{c}_1[n]\mathrm{c}_2[n]P_t^2[n] + 2\mathrm{c}_1[n]\mathrm{c}_2[n]P_t[n]\widetilde{\sigma}_{n_u}^2 + 2\mathrm{c}_1[n]P_t[n]\widetilde{\sigma}_{n_u}^2 + \mathrm{c}_1[n]\widetilde{\sigma}_{n_u}^4 + \widetilde{\sigma}_{n_u}^2}{\ln(2)(1+\mathrm{c}_1[n]P_t[n])\left(\mathrm{c}_2[n]P_t[n] + \widetilde{\sigma}_{n_u}^2\right)\left(\mathrm{c}_2[n]P_t[n] + P_t[n] + \widetilde{\sigma}_{n_u}^2\right)}. \tag{18}$$

Subsequently, by taking the partial derivative of the Lagrangian function with respect to $P_t[n]$ for the $n$th slot, given by (18), then solving $\frac{\partial \mathcal{L}}{\partial P_t[n]} = 0$, three roots of the cubic equation in terms of $\lambda_T$, $\chi_1$, $\chi_2$, and $\chi_3$ are obtained, respectively. Among these three solutions, each root can be written explicitly via the cubic formula (see Appendix C). We then select the real positive solutions that yield the highest $\widetilde{R}[n]$, expressed as

$$P_t^*[n] = \underset{P_t[n] \in \{\chi_1, \chi_2, \chi_3\} \cap \mathbb{R}^+}{\operatorname{argmax}} \widetilde{R}[n] \tag{19}$$

Thus, the total transmitted power $P_{tot} = \sum_{n=1}^{N} P_t^*[n]$ is obtained by summing all the power $P_t[n]$ for all slots. To ensure at least one feasible solution from (19) in the $n$th slot, we need to check $\lambda_T$ at the boundary conditions. By utilizing the GSS method, the largest $\lambda_T$ can be selected as the upper bound by setting $P_t[n] = 0$ in the $n$th slot and substituting it into $\frac{\partial \mathcal{L}}{\partial P_t[n]} = 0$. Consequently, the range of $\lambda_T$ is restricted to the following interval:

$$0 \leq \lambda_T \leq \lambda_{max}, \text{ where } \lambda_{max} \triangleq \max_{\forall n \in \mathcal{N}} \left\{ \frac{\mathrm{c}_1[n]\,\widetilde{\sigma}_{n_u}^2 + 1}{\ln(2)\,\widetilde{\sigma}_{n_u}^2\, t_{hov}} \right\}. \tag{20}$$

Next, the optimal Lagrange multiplier $\lambda_T^*$ with the corresponding transmitted power $P_t^*[n]$ for each time slot can be determined using the GSS algorithm [35], as outlined in Algorithm 1. Eventually, recalling from (13), the maximum throughput is achieved.

**Algorithm 1** Computation of Optimal Power Allocation $P_t^*[n]$

**Input:** Convergence threshold $\varepsilon$, GSS tolerance $\epsilon$, total served BNs $K$, constants $\widetilde{\sigma}_{n_u}^2$, $c_1[n]$, $c_2[n]$, collection location $(x_u[n], y_u[n])$, lateral distance $r[n]$, reflection coefficients $\Gamma_i[n]$, and channel gain $h_i[n]$, $\forall i \in \{1, 2\}$.
**Output:** Optimal $\lambda_T^*$ and $P_t^*[n]$.

1: Compute and set the lower bound threshold $\mathcal{X}_L = 0$ and the upper bound threshold $\mathcal{X}_H = \lambda_{max}$ using (20).
2: Set ctr $= 1$ and $\lambda_T^{(1)} \leftarrow (\mathcal{X}_L + \mathcal{X}_H)/2$.
3: Obtain the transmit power $P_t^{(1)}[n]$ from (19) by substituting $\lambda_T^{(1)}$ into (18) and solving the cubic equation.
4: Substitute $P_t^{(1)}[n]$ into (16). Set $R_s^{(1)}[n] \leftarrow \widetilde{R}[n]$.
5: **repeat** (Main Loop) ▷ Iteration
6: Update ctr $=$ ctr $+ 1$.
7: For $\lambda_T^{(\text{ctr})} \in [\mathcal{X}_L, \mathcal{X}_H]$, using golden search within an acceptable tolerance $\epsilon$, compute the optimal solution $\lambda_T^{(\text{ctr})}$ with $P_t^{(\text{ctr}-1)}[n]$ to maximize $\widetilde{R}[n]$.
8: Obtain the transmit power $P_t^{(\text{ctr})}[n]$ from (19) by substituting $\lambda_T^{(\text{ctr})}$ into (18) and solving the cubic equation.
9: Substitute $P_t^{(\text{ctr})}[n]$ into (16), set their sum as the sum-throughput rate to $R_s^{(\text{ctr})}[n]$.
10: **until** $\left| R_s^{(\text{ctr})}[n] - R_s^{(\text{ctr}-1)}[n] \right| < \varepsilon$ ▷ Termination
11: **return** Optimal Lagrange multiplier $\{\lambda_T^* \leftarrow \lambda_T^{(\text{ctr})}\}$ and power $\{P_t^*[n] \leftarrow P_t^{(\text{ctr})}[n]\}$.

### *4.4. Optimal Collection Location Design*

Based on the previous section, we obtained the optimal UAV trajectory by optimally grouping the pairs of BNs. In addition, we obtained the optimal backscattering reflection coefficients for each $i$th BN in the $n$th slot. Further, we determined the optimal power allocation in each time slot. Now we need to choose the optimal collection points $(x_u[n], y_u[n])$ for the UAV. Recall that the lateral distance between the UAV and the first BN in $n$th slot can be denoted by $r[n]$ and could be expanded to $r[n] = \sqrt{(x_u[n] - x_1[n])^2 + (y_u[n] - y_1[n])^2}$. Subsequently, we can determine the optimal collection point $(x_u^*[n], y_u^*[n])$ in each time slot by obtaining the optimal $r^*[n]$. In real-life scenarios, the coherence time of the UAV is short, where the channel gain remains constant [36]. Meanwhile, the UAV could operate at a relatively high altitude to maintain a reachable LoS path. Here, we propose a practically motivated approximation for the convexity check of the collection location optimization. For analytical tractability, it is assumed that the UAV operates at a sufficiently high altitude $z$, that is, $z \gg r[n]$.

**Remark 1.** *The assumption $z \gg r[n]$ is invoked solely to establish convexity and derive the closed-form solution for $\mathcal{P}_{D1}$. All numerical results are evaluated using the exact path loss model in (5). Empirical analysis of the considered deployment scenarios confirms that the average inter-node distance is consistently below z/10 m, and over 73% of BN pairs satisfy $z > 10 \cdot r[n]$, validating this approximation in practice.*

As a result, when the elevation angle $\phi_i[n]$ is assumed to be $90^\circ$ at all slots, we have $\sin^{-1}\left(\frac{z}{d_i[n]}\right) \approx \pi/2$, which leads to $\phi_i[n] \approx 90^\circ$ in (3). Under this condition, the wireless channel links between the UAV and $K$ BNs can predominantly be regarded as LoS paths. Consequently, the average path loss in (5) can be expressed in terms of $d_i[n]$ as

$$\frac{\partial \widehat{R}[n]}{\partial r[n]} = r[n]^2\left(r[n]^2+z^2\right)^{-\frac{3}{2}-\Gamma_1[n]}\left(2e^{2b\frac{180}{\pi}\sin^{-1}\left(\frac{z}{\sqrt{r[n]^2+z^2}}\right)}\right)\left(\sqrt{r[n]^2+z^2}\Gamma_1[n]\beta_0\right)\left(ae^{ab}+e^{b\frac{180}{\pi}\sin^{-1}\left(\frac{z}{\sqrt{r[n]^2+z^2}}\right)}\right)^{-2}$$
$$\times\left(-\sqrt{\frac{r[n]^2}{r[n]^2+z^2}}\right)^{-1}+2a^2\Gamma_1[n]\kappa\beta_0 e^{2ab}r[n]+ae^{b\left[a+\frac{180}{\pi}\sin^{-1}\left(\frac{z}{\sqrt{r[n]^2+z^2}}\right)\right]}\left(\frac{180}{\pi}bz(\beta_0-\kappa\beta_0)+2\Gamma_1[n]r[n](\beta_0+\kappa\beta_0)\right). \quad (21)$$

$$\begin{aligned} L_i[n] &= \frac{\beta_0\mathcal{P}_i^{LoS}[n]}{d_i^2[n]}+\frac{\kappa\beta_0\left(1-\mathcal{P}_i^{LoS}[n]\right)}{d_i^2[n]}, \\ &= \frac{\beta_0\mathcal{P}_i^{LoS}[n](1-\kappa)+\kappa\beta_0}{d_i^2[n]}, \quad \forall i\in\{1,2\}. \end{aligned} \quad (22)$$

Subsequently, by letting other optimization variables be fixed, the total throughput $\widehat{R}[n]$ at the $n$th slot can be rewritten with respect to $d_i[n]$ as follows:

$$\widehat{R}[n] \triangleq \log_2\left(1+\frac{\mathrm{e}_1[n]}{d_1^4[n]}\right)+\log_2\left(1+\frac{\frac{\mathrm{e}_2[n]}{d_2^4[n]}}{\frac{\mathrm{e}_1[n]}{d_1^4[n]}+\sigma_{n_u}^2}\right). \quad (23)$$

Here, we first define $\widehat{L}_i[n] \triangleq \beta_0\mathcal{P}_i^{LoS}[n](1-\kappa)+\kappa\,\beta_0,\ \forall i\in\{1,2\}$. Further, we denote $\mathrm{e}_1[n] \triangleq \frac{\Gamma_1[n]P_t[n]|h_1[n]|^4\left|\widehat{L}_1[n]\right|^2}{\sigma_{n_u}^2}$, and $\mathrm{e}_2[n] \triangleq \Gamma_2[n]P_t[n]|h_2[n]|^4\left|\widehat{L}_2[n]\right|^2$. Following this reformulation, the optimization problem $\mathcal{O}_\mathrm{S}$ can be reduced to the following subproblem $\mathcal{P}_{\mathrm{D1}}$, expressed as

$$\mathcal{P}_{\mathrm{D1}}: \max_{\{r[n],\forall n\in\mathcal{N}\}} \widehat{R}_{\mathrm{Sum}}=\sum_{n=1}^{N}\widehat{R}[n], \quad \text{subject to:} \quad (\mathrm{C3}).$$

**Lemma 3.** *Problem $\mathcal{P}_{\mathrm{D1}}$ is a convex optimization problem.*

**Proof.** The proof is provided in Appendix D. □

Since $\mathcal{P}_{\mathrm{D1}}$ is convex while (C3) is a linear constraint, the globally optimized solution of $r^*[n]$ can be obtained by taking the derivative of $\widehat{R}[n]$ at the $n$th slot with respect to $r[n]$, as shown in (21), then by solving $\frac{\partial\widehat{R}[n]}{\partial r[n]}=0$, which yields roots $\bar{r}[n]$. Therefore, the globally optimized lateral distance $r^*[n]$ is obtained by selecting the best in terms of the maximum sum-throughput rate among the lower or upper boundary points $\{0,\, d_{N2N}\}$, or $\bar{r}[n]$, that satisfies the constraint (C3). Subsequently, the new optimal collection point is derived by calculating the optimal angle $\theta^*[n]$ formed between the line segment that connects two BNs and the x-axis at the $n$th slot:

$$\theta^*[n] = \sin^{-1}\left(\frac{y_2[n]-y_1[n]}{d_{N2N}[n]}\right). \quad (24)$$

Subsequently, the optimal collection point $(x_u^*[n], y_u^*[n])$ in the $n$th slot is determined as follows:

$$x_u^*[n] = x_1[n] + kr^*[n]\cos\theta^*[n], \quad \text{for } k = \begin{cases} -1, & x_1[n] > x_2[n] \\ 1, & \text{Otherwise} \end{cases}$$
$$y_u^*[n] = y_1[n] - r^*[n]\sin\theta^*[n]. \tag{25}$$

### *4.5. Jointly Optimal Design*

The optimal solution to $\mathcal{O}_{\mathrm{S}}$ is obtained by iteratively solving subproblems $\mathcal{P}_{\mathrm{A}}$, $\mathcal{P}_{\mathrm{B1}}$, $\mathcal{P}_{\mathrm{B2}}$, $\mathcal{P}_{\mathrm{C1}}$, and $\mathcal{P}_{\mathrm{D1}}$, respectively. The corresponding pseudo-code is summarized in Algorithm 2. Since each subproblem can be optimally solved by updating only one variable while keeping all other variables fixed, the sum-throughput rate $R_{\mathrm{Sum}}$ is expected to increase monotonically and converge to a stationary value within a few iterations. (Among the four subproblems, $\mathcal{P}_{B1}$ and $\mathcal{P}_{B2}$, $\mathcal{P}_{C1}$, and $\mathcal{P}_{D1}$ are each solved with provable global optimality, while the trajectory subproblem $\mathcal{P}_A$ is addressed via a GA-based heuristic due to the NP-hard nature of the TSP [31]. An exhaustive search-based upper bound is computationally infeasible for the considered problem sizes. Nevertheless, the monotonic convergence of Algorithm 2 and the significant performance gains over benchmark schemes provide strong evidence of near-optimal performance.) To verify the effectiveness and convergence behavior of the proposed joint design, numerical results are presented in Section 5 and illustrated in Figure 2. These results serve as evidence to demonstrate that the algorithm successfully converges to a feasible solution, making it a reliable approach for solving the optimization problem.

**Algorithm 2** Joint Optimization of Solving $\mathcal{O}_{\mathrm{S}}$ to find the Maximum Sum-Throughput Rate $R_{\mathrm{Sum}}$ by Iterative Algorithm

**Input:** UAV starting point $\mathbf{U_S}(x_S, y_S, z)$ and ending point $\mathbf{U_E}(x_E, y_E, z)$, convergence threshold $\xi$, power budget $P_b$, UAV altitude $z$, total served BNs $K$, total slots $N$, noise power $\sigma_{n_u}^2$, channel gain $h_i[n]$, $\forall i \in \{1, 2\}$.

**Output:** Optimal reflection coefficients $\Gamma_1^*[n]$, and $\Gamma_2^*[n]$, trajectory routes $\mathcal{T}^*$, collection location $(x_u^*[n], y_u^*[n])$, power allocation $P_t^*[n]$, maximum sum-throughput $R_{\mathrm{Sum}}^*$.

1: Initialize ctr $= 1$, and $P_t^{(1)}[n] = \frac{1}{2}\frac{P_b}{N}$, $\forall n \in \mathcal{N}$.
2: Initialize $\Gamma_1^{(1)}[n] = \Gamma_2^{(1)}[n] = 0.5$, $\forall n \in \mathcal{N}$.
3: Set $\left(x_u^{(1)}[n], y_u^{(1)}[n]\right) = \left(\frac{(x_1[n]+x_2[n])}{2}, \frac{(y_1[n]+y_2[n])}{2}\right), \forall n$.
4: **repeat** (Main Loop) ▷ Iteration
5: Update ctr $=$ ctr $+ 1$.
6: Obtain the optimal trajectory routes $\{\mathcal{T}^{(\mathrm{ctr}-1)}\}$ by solving $\mathcal{P}_{\mathrm{A}}$, with given power $\{P_t^{(\mathrm{ctr}-1)}[n]\}$, reflection coefficients $\{\Gamma_1^{(\mathrm{ctr}-1)}[n]\}$, and $\{\Gamma_2^{(\mathrm{ctr}-1)}[n]\}$, and the collection location $\{(x_u^{(\mathrm{ctr}-1)}[n], y_u^{(\mathrm{ctr}-1)}[n])\}$.
7: Obtain the optimal closed-form solutions of $\{\Gamma_1^{(\mathrm{ctr})}[n]\}$, $\{\Gamma_2^{(\mathrm{ctr})}[n]\}$ by solving $\mathcal{P}_{\mathrm{B2}}$ and $\mathcal{P}_{\mathrm{B1}}$ respectively, given fixed $\{\mathcal{T}^{(\mathrm{ctr}-1)}\}$, $\{P_t^{(\mathrm{ctr}-1)}[n]\}$, and $\{(x_u^{(\mathrm{ctr}-1)}[n], y_u^{(\mathrm{ctr}-1)}[n])\}$.
8: Obtain the optimal solution to $\{P_t^{(\mathrm{ctr})}[n]\}$ by solving $\mathcal{P}_{\mathrm{C1}}$ via the procedure in Algorithm 1, with fixed parameters of $\{\Gamma_1^{(\mathrm{ctr})}[n]\}$, $\{\Gamma_2^{(\mathrm{ctr})}[n]\}$, $\{\mathcal{T}^{(\mathrm{ctr}-1)}\}$ and $\{(x_u^{(\mathrm{ctr}-1)}[n], y_u^{(\mathrm{ctr}-1)}[n])\}$.
9: Obtain the optimal solution of $\{(x_u^{(\mathrm{ctr})}[n], y_u^{(\mathrm{ctr})}[n])\}$ by solving $\mathcal{P}_{\mathrm{D1}}$, under given $\{\Gamma_1^{(\mathrm{ctr})}[n]\}$, $\{\Gamma_2^{(\mathrm{ctr})}[n]\}$, $\{\mathcal{T}^{(\mathrm{ctr}-1)}\}$, and $\{P_t^{(\mathrm{ctr})}[n]\}$.
10: **until** $\left|R_{\mathrm{Sum}}^{(\mathrm{ctr})} - R_{\mathrm{Sum}}^{(\mathrm{ctr}-1)}\right| < \xi$ ▷ Termination
11: **return** $\{\mathcal{T}^* \leftarrow \mathcal{T}^{(\mathrm{ctr}-1)}\}$, $\{\Gamma_1^*[n] \leftarrow \Gamma_1^{(\mathrm{ctr})}[n]\}$, $\{\Gamma_2^*[n] \leftarrow \Gamma_2^{(\mathrm{ctr})}[n]\}$, $\{(x_u^*[n], y_u^*[n]) \leftarrow (x_u^{(\mathrm{ctr})}[n], y_u^{(\mathrm{ctr})}[n])\}$, $\{P_t^*[n] \leftarrow P_t^{(\mathrm{ctr})}[n]\}$, and $\{R_{\mathrm{Sum}}^* \leftarrow R_{\mathrm{Sum}}^{(\mathrm{ctr})}\}$.

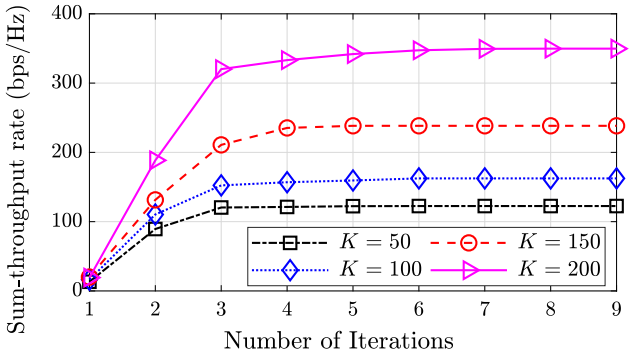



**Figure 2.** Fast convergence of the proposed Algorithm 2.

### *4.6. Complexity Analysis and Runtime Evaluation*

The computational time complexity of the proposed joint design in Algorithm 2 is analyzed to evaluate the efficiency of the algorithm concerning the amount of data to be processed. In general, this complexity is characterized as a function of the input data size and is governed by the number of variables and constraints involved in the optimization problem. Specifically, the optimization problem $\mathcal{O}_{\mathrm{S}}$ is decomposed into four subproblems, each addressed within a single iteration of Algorithm 2. We let $N$ represent the total number of time slots. We begin by analyzing the computational complexity of each subproblem individually.

First, for the trajectory optimization subproblem $\mathcal{P}_{\mathrm{A}}$ which is formulated as a TSP, the worst-case computational complexity of an exact solution is $\mathcal{O}((N+1)!)$ due to the NP-hard nature of the problem [31]. In this work, a GA-based heuristic is employed to obtain near-optimal solutions at a significantly reduced computational cost; nevertheless, we retain the worst case factorial complexity for a conservative analysis. Moreover, the optimal backscattering reflection coefficients are obtained by solving $\mathcal{P}_{\mathrm{B1}}$ and $\mathcal{P}_{\mathrm{B2}}$. Since each pair of BNs' reflection coefficients can be optimized individually in closed form for each time slot, the total complexity of solving both subproblems is $\mathcal{O}(N)$. Furthermore, Algorithm 1 is utilized to find the optimal power allocation by solving $\mathcal{P}_{\mathrm{C1}}$. The optimal solution is achieved within an acceptable tolerance $\epsilon$ applying 1D GSS [35] in the range $[\mathcal{X}_L, \mathcal{X}_H]$. Here, we denote the lower bound $\mathcal{X}_L = 0$ and the upper bound $\mathcal{X}_H = \max_{\forall n \in \mathcal{N}} \{\frac{\mathrm{c}_1[n]\,\tilde{\sigma}^2_{n_u}+1}{\ln(2)\,\tilde{\sigma}^2_{n_u}\,t_{hov}}\}$. In each iteration of the GSS, the cubic equation in (18) is solved for all $N$ slots using the closed-form Cardano formula derived in Appendix C, which requires $\mathcal{O}(1)$ per slot. An additional $\mathcal{O}(1)$ comparison step is needed to select the appropriate real positive root from (19). By letting $N_{\mathrm{itr}}$ denote the number of GSS iterations required for Algorithm 1 to converge, the search space diminishes by a factor of 0.618 in each iteration and terminates when $\Delta\mathcal{X}_S(0.618)^{N_{\mathrm{itr}}} \leq \epsilon$, yielding $N_{\mathrm{itr}} \leq 2\ln\left(\frac{\mathcal{X}_H}{\epsilon}\right)$, where $\Delta\mathcal{X}_{\mathcal{S}} \triangleq \mathcal{X}_H - \mathcal{X}_L = \mathcal{X}_H$. Consequently, the total complexity of $\mathcal{P}_{\mathrm{C1}}$ is $\mathcal{O}\left(N\ln\left(\frac{\mathcal{X}_H}{\epsilon}\right)\right)$. Finally, the collection location subproblem $\mathcal{P}_{\mathrm{D1}}$ admits a closed-form solution per slot, resulting in a total complexity of $\mathcal{O}(N)$. Overall, by denoting $M_{\mathrm{itr}}$ as the number of iterations required for the convergence of Algorithm 2, the total computational complexity of the joint design is given by $\mathcal{O}\Big(M_{\mathrm{itr}}\Big((N+1)! + N\ln\Big(\frac{\mathcal{X}_H}{\epsilon}\Big) + 3N\Big)\Big) \approx \mathcal{O}(M_{\mathrm{itr}}(N+1)!)$, for serving large-scale BNs, which correspond to a larger number of time slots for data collection.

To complement the above theoretical complexity analysis, we also evaluate the actual computational runtime of the proposed joint design. All simulations are conducted in MATLAB R2024b on a Dell Precision 7780 workstation with a 13th Generation Intel Core i9-13950HX processor and 64 GB of RAM. Table 3 summarizes the averaged runtime over multiple independent trials for varying numbers of served BNs $K = \{50, 100, 150, 200\}$. The results show that the total runtime scales reasonably with $K$, increasing from approximately

14 s for $K = 50$ to 88 s for $K = 200$. The trajectory optimization subproblem $\mathcal{P}_{\mathrm{A}}$ dominates the computation due to its combinatorial nature, accounting for over 90% of the total runtime across all cases. In contrast, the reflection coefficient and collection location subproblems via $\mathcal{P}_{\mathrm{B1}}$, $\mathcal{P}_{\mathrm{B2}}$, and $\mathcal{P}_{\mathrm{D1}}$ require negligible computation owing to their closed-form solutions, while the power allocation subproblem $\mathcal{P}_{\mathrm{C1}}$ accounts for a moderate share that grows proportionally with $K$. These results confirm that the proposed joint design remains computationally tractable for practical offline UAV mission planning.

**Table 3.** Averaged computational runtime evaluation under different values of $K$.

| $K$ | Total (s) * | Subproblem $\mathcal{P}_{\mathrm{A}}$ (s) | Subproblem $\mathcal{P}_{\mathrm{B1}}$ and $\mathcal{P}_{\mathrm{B2}}$ (s) | Subproblem $\mathcal{P}_{D1}$ (s) | Subproblem $\mathcal{P}_{C1}$ (s) | Iterations |
|---|---|---|---|---|---|---|
| 50 | 14.27 | 13.44 | 0.35 | 0.02 | 0.45 | 2 |
| 100 | 26.09 | 24.62 | 0.34 | 0.01 | 1.12 | 3 |
| 150 | 42.45 | 39.71 | 0.48 | 0.02 | 2.25 | 4 |
| 200 | 88.12 | 82.13 | 0.63 | 0.04 | 5.31 | 7 |

Note *: All runtime values are reported in seconds (s).

## 5. Numerical Results and Performance Evaluation

This section presents the simulation results utilized to evaluate the performance of the proposed joint design. The following numerical results are obtained through Monte Carlo simulations based on the exact system model and serve to validate the analytical claims and the effectiveness of the proposed optimization framework. Unless stated otherwise, the simulation parameters are set as follows: $G_t = G_r = 1$, $\sigma^2_{n_u} = -175$ dBm/Hz, $f_c = 915$ MHz, $\kappa = 0.3$, $z = 50$ m, $t_{hov} = 100$ ms, $\vartheta_u = 15.521$ m/s, which are adopted from the quad-rotor UAV specifications in [30], and the constants $a, b$ are set to 12.08 and 0.11 [26,27], respectively. In particular, the UAV's battery utilization is modeled with practical consideration, including the battery's energy density relative to its weight and size, as well as the material composition of the battery [37], such as lithium-ion. Accordingly, the maximum energy budget is set to $E_{max} = 1.7 \times 10^4$ J [37]. Additionally, the propulsion power of the UAV is calculated as 108.9118 Watts, based on (11), with the corresponding variables detailed in Table 2.

### *5.1. Validation and Insights of the Joint Design*

#### 5.1.1. Convergence Check

On one hand, Figure 2 illustrates the sum-throughput rate $R_{\mathrm{Sum}}$ against the number of iterations for four scenarios, with the number of BNs set to $K = \{50, 100, 150, 200\}$. The figure shows that the sum rate $R_{\mathrm{Sum}}$ increases iteratively and converges after a few iterations, specifically at the $\{4, 5, 6, 8\}$th iteration. This demonstrates that serving more BNs requires a longer time to reach convergence due to the increased computational complexity introduced by a larger number of BNs, which demand more computational resources and time for optimization. Additionally, note that the final converged throughput values are distinct for each $K$, confirming that the system performance scales with the number of served BNs. Moreover, the rapid increase in throughput at the initial iterations flattens as the number of iterations grows, indicating that the system is approaching the optimal solution, with diminishing returns in terms of throughput improvement in subsequent iterations.

On the other hand, Figure 3 shows the sum-throughput rate $R_{\mathrm{Sum}}$ against the number of served BNs under two different area sizes, $100 \times 100\,\mathrm{m}^2$ and $200 \times 200\,\mathrm{m}^2$, and two fixed UAV altitude settings, which are at 50 m and 75 m. As expected, the overall $R_{\mathrm{Sum}}$ increases with the number of served BNs, reflecting the capacity of the system to handle massive

and large-scale BN deployments. However, this growth rate slows for larger areas and higher UAV altitudes due to increased energy consumption along longer trajectories and greater path loss at higher altitudes. In particular, the sum-throughput rate is lower in the larger $200 \times 200\,\mathrm{m}^2$ area compared to the $100 \times 100\,\mathrm{m}^2$ area. Similarly, the sum-throughput rate decreases as the UAV altitude increases from 50 m to 75 m. These results underscore the need to balance area size and altitude, as both impose limitations due to energy constraints and increased path loss, highlighting the importance of optimizing the UAV's operational parameters. Meanwhile, we note that the stabilized throughput values differ across all four configurations, confirming the distinct impact of area size and altitude on system performance.

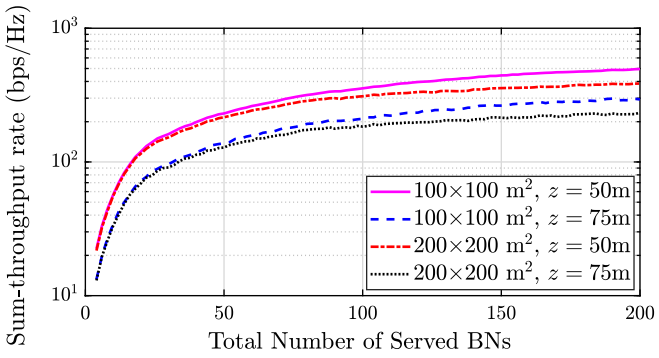



**Figure 3.** Sum-throughput rate for different UAV flight area sizes and altitudes against varying numbers of served BNs.

5.1.2. Trajectory Planning and Optimal Collection Points

Following the discussion in Section 4.1, we implement the GA for the trajectory optimization problem. Figure 4 illustrates the optimal trajectory routes of a UAV for two scenarios with different numbers of time slots $N = \{100, 200\}$ in an area of $200 \times 200\,\mathrm{m}^2$. It can be observed that as $N$ increases, the total traveled distance grows accordingly. Consequently, more slots indicate that the UAV visits more collection points, resulting in a denser trajectory with more flexible route planning. This indicates that increasing $N$ facilitates more extensive data collection from the collection points, potentially enhancing the overall $R_{\mathrm{Sum}}$ performance. In addition, Figure 5 depicts the per-slot throughput as a function of the normalized lateral distance $r[n]$ for four representative time slots, namely the $\{25, 50, 75, 100\}$th slots out of $N$ total slots. In each subplot, the red star marks the optimal UAV collection position in the 2D plane between the paired BNs, at which the per-slot throughput is maximized. Moreover, the uniqueness of the global optimum in each slot is guaranteed by the convexity established in Lemma 3.

**Remark 2.** *The optimization results indicate a single global optimal collection location for the UAV, which can effectively minimize collection time and optimize resource allocation.*

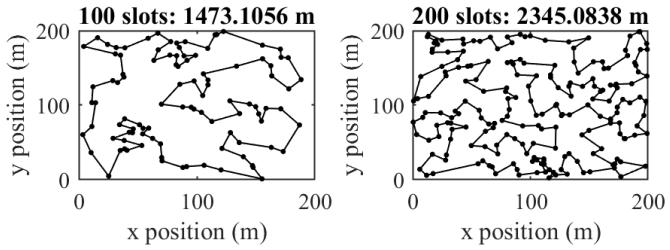



**Figure 4.** Optimal trajectory routes for various collection points.

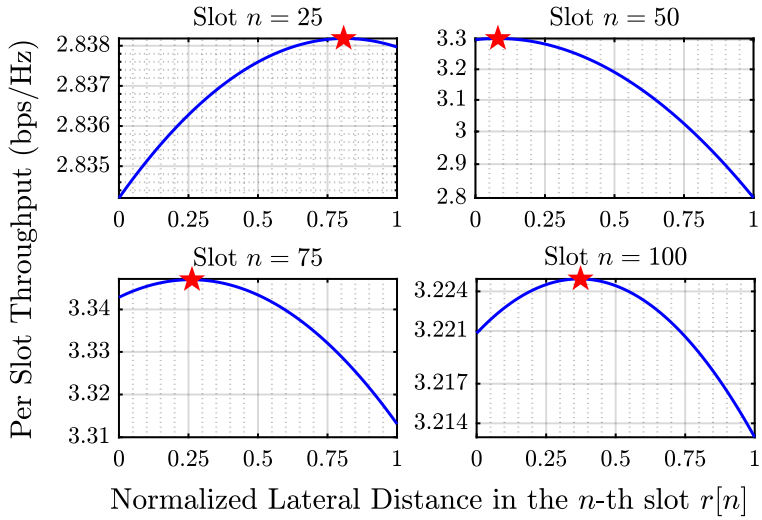



**Figure 5.** Optimal collection point in four different slots. The red star marks the optimal point along the lateral distance that maximizes the per-slot throughput.

#### 5.1.3. Power Allocation Analysis

Here, Figure 6 depicts the total transmit energy consumption for various numbers of the served BNs, $K = \{50, 100, 150, 200\}$, as a function of the Lagrange multiplier. Here, we denote the terms $\lambda_{T,100}$ and $\lambda_{T,200}$ corresponding to two UAV flying areas of $100 \times 100\,\mathrm{m}^2$ and $200 \times 200\,\mathrm{m}^2$, respectively. At small values of $\lambda$, a larger $K$ consistently yields higher total transmit energy, as serving more BNs naturally demands greater power resources. However, for each value of $K$, the proposed power allocation scheme effectively identifies an optimal Lagrange multiplier via the 1D GSS within the boundary $[\mathcal{X}_L, \mathcal{X}_H]$ that minimizes the total transmit energy consumption. Notably, this minimization capability remains consistent as the flying area increases from $100 \times 100\,\mathrm{m}^2$ to $200 \times 200\,\mathrm{m}^2$, demonstrating the robustness of the proposed scheme across different deployment scales.

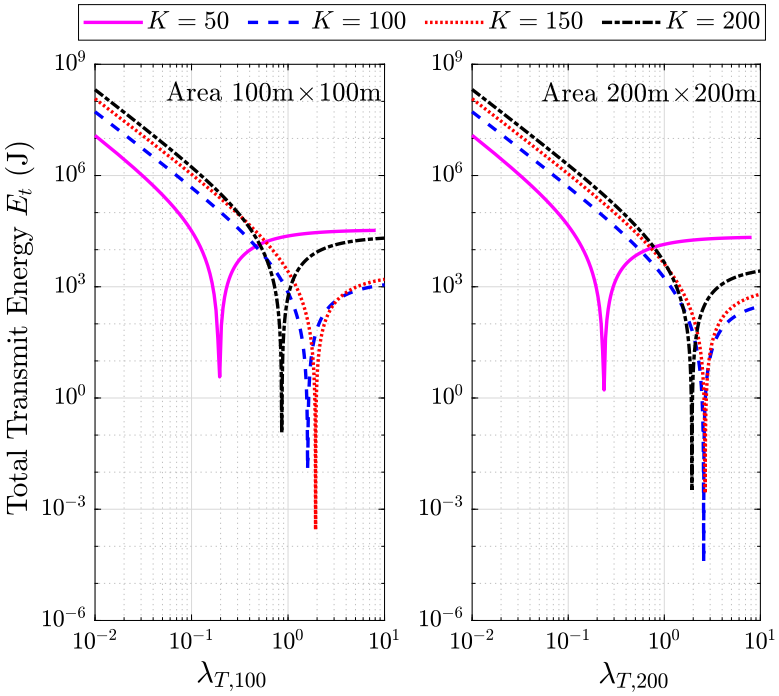



**Figure 6.** Optimal power allocation for two different area sizes.

**Remark 3.** *The Lagrange Multiplier with one optimal $\lambda$ for all slots signifies the convexity of the power allocation scheme, which can effectively maximize power efficiency.*

### *5.2. Performance Comparison and Gains*

Here, we compare the sum-throughput performance against the relevant benchmarks. (A direct numerical comparison with existing works is not discussed and provided because most adopt different system models, objectives, and parameter settings, making a fair evaluation under identical conditions infeasible. Instead, benchmark schemes constructed under the same assumptions are used for consistency and fairness.) Specifically, our proposed joint design is denoted as "Proposed", which optimizes all variables, including UAV's trajectory, backscattering reflection coefficients, power allocation, and the collection locations, i.e., $\{\mathcal{T}^*, \Gamma_1^*[n], P_t^*[n], (x_u^*[n], y_u^*[n])\}$, respectively, with the benchmarks shown in Figure 7, namely: (1) Optimization of only the reflection coefficient $\Gamma_1[n]$, denoted as "Optimal $\Gamma_1[n]$"; (2) Optimization of only the collection locations $(x_u^*[n], y_u^*[n])$, equivalent to optimizing the lateral distance $r[n]$, denoted as "Optimal $r[n]$"; and (3) Optimization of only the power allocation $P_t[n]$, which denotes as "Optimal $P_t[n]$". The results clearly show that the proposed joint design consistently outperforms the other benchmark schemes, especially as the number of served BNs increases. Notably, as the number of served BNs reaches 200, the joint optimization scheme exhibits a substantial advantage, maintaining the highest sum-throughput rate across all variations. The performance gains are further quantified in Figure 8, which illustrates the enhancement of each individual optimization strategy and the proposed joint design compared to the fixed scheme. Power allocation optimization yields the highest individual improvement, with a performance enhancement of 8.295 dB, making it the most influential factor in improving overall throughput. Trajectory optimization follows, contributing a 2.519 dB improvement, while reflection coefficient optimization provides a moderate gain of 41.4%. Collection point selection offers a minimal improvement of 1.9%. Overall, when these optimizations are combined in the joint design, the performance enhancement reaches 8.983 dB, emphasizing the importance of integrating multiple optimization strategies to achieve the best possible system performance. Both figures underscore the significant advantages of the joint optimization approach over individual strategies. The combined effect of these strategies offers a comprehensive solution for maximizing system performance, with the joint optimization achieving the highest overall gains.

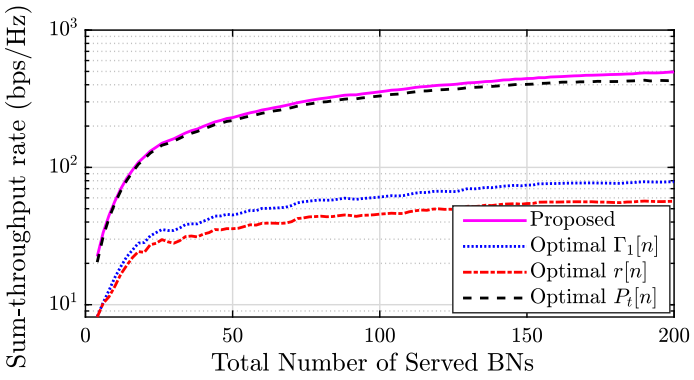



**Figure 7.** Sum-throughput performance comparison of the proposed design against benchmarks under different variations.

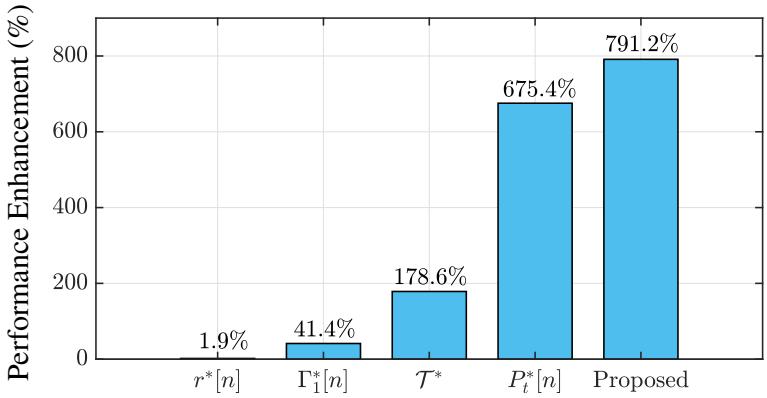



**Figure 8.** Performance enhancement of the proposed joint design against benchmarks under different variations.

## 6. Concluding Remarks and Future Directions

In this paper, we propose an optimization framework to maximize the sum-throughput rate of a UAV-assisted data transmission network. To achieve this, the problem is decomposed into four stages, each of which is solved independently. Real-world noise is accounted for by calculating the reflection coefficient using both BackCom and NOMA schemes, which leads to significant improvements in system performance. Energy conservation is achieved through optimal UAV trajectory selection, while an efficient power allocation scheme contributes to overall system efficiency. Simulation results clearly demonstrate the effectiveness of our methodology, showing a sum-throughput increase of 8.983 dB compared to the benchmark schemes. From a practical perspective, the proposed framework is applicable to a range of UAV-assisted IoT deployment scenarios, including precision agriculture, environmental monitoring, smart logistics with RFID-based inventory management, and post-disaster emergency data collection, where energy-efficient and scalable data collection from battery-less ground devices is essential. The adopted UAV parameters are consistent with commercially available lightweight quad-rotor platforms, further supporting the practical relevance of the proposed design. Future work may extend the current two-user NOMA pairing scheme to a multi-user NOMA configuration to serve more BNs simultaneously, potentially improving spectral efficiency in ultra-dense IoT deployments. Furthermore, extending the design to 3D trajectory optimization with adaptive altitude control could further enhance channel conditions and energy efficiency, particularly in complex urban environments.

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**Conflicts of Interest:** The authors declare no conflicts of interest.

## Appendix A. Finding the Critical Points for $\overline{R}[n]$ in $\Gamma_1[n]$

In this section, we provide the detailed derivative of $\overline{R}[n]$ with respect to $\Gamma_1[n]$ in the $n$th slot. We first redefine the total backscattered throughput $\overline{R}[n]$ recalling from (8) and (9), as

$$\begin{aligned}\overline{R}[n] &= \log_2(1+\Gamma_1[n]\mathrm{b}_2[n]) + \log_2\left(1+\frac{1}{\Gamma_1[n]\mathrm{b}_3[n]+\overline{\sigma}^2_{n_u}}\right),\\ &= \log_2\left[(1+\Gamma_1[n]\mathrm{b}_2[n])\left(1+\frac{1}{\Gamma_1[n]\mathrm{b}_3[n]+\overline{\sigma}^2_{n_u}}\right)\right].\end{aligned} \tag{A1}$$

Subsequently, by taking the first derivative of $\overline{R}[n]$ with respect to $\Gamma_1[n]$ and setting it to 0, all critical points are found to maximize $\overline{R}[n]$:

$$\begin{aligned}\frac{\partial \overline{R}[n]}{\partial \Gamma_1[n]} &= 0,\\ \frac{\mathrm{b}_2[n]\mathrm{b}_3{}^2[n]\Gamma_1^2[n]+\mathrm{b}_2[n]\overline{\sigma}^2_{n_u}\left(1+\overline{\sigma}^2_{n_u}\right)-\mathrm{b}_3[n]\left(1-2\mathrm{b}_2[n]\Gamma_1[n]\overline{\sigma}^2_{n_u}\right)}{\ln(2)(1+\mathrm{b}_2[n]\Gamma_1[n])\left(\mathrm{b}_3[n]\Gamma_1[n]+\overline{\sigma}^2_{n_u}\right)\left(1+\mathrm{b}_3[n]\Gamma_1[n]+\overline{\sigma}^2_{n_u}\right)} &= 0,\\ \mathrm{b}_2[n]\mathrm{b}_3{}^2[n]\Gamma_1^2[n]+\mathrm{b}_2[n]\overline{\sigma}^2_{n_u}\left(1+\overline{\sigma}^2_{n_u}\right)-\mathrm{b}_3[n]\left(1-2\mathrm{b}_2[n]\Gamma_1[n]\overline{\sigma}^2_{n_u}\right) &= 0.\end{aligned} \tag{A2}$$

By solving the above equation, all the critical points of $\overline{\Gamma}_1[n]$ for maximizing $\overline{R}[n]$ are obtained and derived as follows:

$$\overline{\Gamma}_1[n] \triangleq \frac{-\mathrm{b}_2[n]\mathrm{b}_3[n]\overline{\sigma}^2_{n_u} \pm \sqrt{-\mathrm{b}_2[n]\mathrm{b}_3{}^2[n]\left(-\mathrm{b}_3[n]+\mathrm{b}_2[n]\overline{\sigma}^2_{n_u}\right)}}{\mathrm{b}_2[n]\mathrm{b}_3{}^2[n]}. \tag{A3}$$

## Appendix B. Proof of Lemma 1: Concavity of $\widetilde{R}[n]$ in $P_t[n]$

Recalling from (16), we can check the concavity of $\widetilde{R}[n]$ with respect to $P_t[n]$ by finding the second derivative, as shown below:

$$\begin{aligned}\frac{\partial^2 \widetilde{R}[n]}{\partial^2 P_t[n]} &= -\frac{\mathrm{c}_1{}^2[n]}{\ln(2)(1+\mathrm{c}_1[n]P_t[n])^2}\\ &\quad - \frac{\widetilde{\sigma}^2_{n_u}\left(\widetilde{\sigma}^2_{n_u}+2\mathrm{c}_2[n]\widetilde{\sigma}^2_{n_u}+2\mathrm{c}_2[n]P_t[n](1+\mathrm{c}_2[n])\right)}{\ln(2)\left(\widetilde{\sigma}^2_{n_u}+\mathrm{c}_2[n]P_t[n]\right)^2\left(\widetilde{\sigma}^2_{n_u}+P_t[n](1+\mathrm{c}_2[n])\right)^2}.\end{aligned} \tag{A4}$$

where $\mathrm{c}_1[n]$, $\mathrm{c}_2[n]$, $\widetilde{\sigma}^2_{n_u}$ are constants, and the transmit power $P_t[n]$ in the $n$th slot is always positive. Therefore, the second derivative $\frac{\partial^2 \widetilde{R}[n]}{\partial^2 P_t[n]} < 0$ indicates that $\widetilde{R}[n]$ is concave with respect to $P_t[n]$. Moreover, since the objective function of $\mathcal{P}_{\mathrm{C1}}$ is concave with respect to $P_t[n]$ and its constraint (C4) is affine, $\mathcal{P}_{\mathrm{C1}}$ constitutes a convex optimization problem. As such, the KKT conditions are sufficient to guarantee global optimality. This concludes the proof of Lemma 1.

## Appendix C. Detailed Steps of Finding KKT Point

By solving $\frac{\partial \mathcal{L}}{\partial P_t[n]} = 0$, we first simplify and re-write (18) as a cubic equation in terms of $t_{hov}$ and $\lambda_T$, with respect to positive real roots $P_t[n]$. Here, the cubic equation can be expressed as

$$AP_t^3[n] + BP_t^2[n] + CP_t[n] + D = 0, \tag{A5}$$

where $A, B, C, D$ are the constant values that can be obtained using the Cardano formula, as shown below, respectively:

$$\begin{cases} A \triangleq -\mathrm{c}_1[n]\mathrm{c}_2[n](1+\mathrm{c}_2[n])t_{hov}\lambda_T \log(2), \\ B \triangleq -\mathrm{c}_2[n](1+\mathrm{c}_2[n])t_{hov}\,\lambda_T \log(2)+\mathrm{c}_2[n](\mathrm{c}_1[n]+\mathrm{c}_2[n]) \\ \qquad -t_{hov}\,\lambda_T\widetilde{\sigma}^2_{n_u}\mathrm{c}_1[n](\log(2)+\mathrm{c}_2[n]\log(4)), \\ C \triangleq -\widetilde{\sigma}^2_{n_u}\mathrm{c}_1[n]\left(-2-2\mathrm{c}_2[n]+t_{hov}\,\lambda_T\,\widetilde{\sigma}^2_{n_u}\log(2)\right) \\ \qquad -\widetilde{\sigma}^2_{n_u}t_{hov}\,\lambda_T(\log(2)+\mathrm{c}_2[n]\log(4)), \\ D \triangleq \widetilde{\sigma}^2_{n_u}+\mathrm{c}_1[n]\,\widetilde{\sigma}^4_{n_u}-t_{hov}\,\lambda_T\,\widetilde{\sigma}^4_{n_u}\log(2). \end{cases}$$

Therefore, we can denote the three roots, $\chi_1, \chi_2, \chi_3$, as the solutions by solving the Cubic Equation (A5).

## Appendix D. Proof of Lemma 3: Concavity of $\widehat{R}[n]$ in $r[n]$

Recalling from (23), we can check the concavity of $\widehat{R}[n]$ with respect to $r[n]$ by finding the second derivative, as shown below:

$$\frac{\partial^2\widehat{R}[n]}{\partial r[n]^2}=\frac{2\left(3r[n]^2-z^2\right)\left(d_{N2N}[n]+2z^2\right)}{\left(r[n]^2+z^2\right)^3}. \tag{A6}$$

Since we assumed in Section 4.4 that the UAV hovers at a relatively high altitude $z$, i.e., $z \gg r[n]$, the term $\left(3r[n]^2-z^2\right)$ is negative for $z \gg r[n]$. Therefore, the second derivative $\frac{\partial^2\widehat{R}[n]}{\partial r[n]^2} < 0$ indicates that $\widehat{R}[n]$ is concave with respect to $r[n]$. Meanwhile, Constraint (C3) is affine. Consequently, $\mathcal{P}_{\mathrm{D1}}$ has a concave objective and convex constraints, and is thus a convex optimization problem. The global optimum $r^*[n]$ is guaranteed. This completes the proof of Lemma 3.

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